

A Hybrid Approach for Job Recommendation Systems

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Abstract—This paper presents a hybrid approach to job recommendation that integrates traditional and Large Language Models to provide more relevant job recommendations. The methodology involves preprocessing CV and job description databases, followed by application processing with two different modules: one using traditional techniques for recommendations and the other with unguided LLMs. The final job recommendation or similarity score is determined by calculating the average of the similarity scores from both modules. This approach is validated by considering some example CVs and job descriptions from the IT field. Our results demonstrate the effectiveness of this hybrid approach for Job Recommendations.

Keywords—Recommendation Systems, Large Language Model (LLM), TF-IDF, Cosine Similarity, CVs, Resume, Jobs, Job Seekers

I. INTRODUCTION

In the evolving landscape of job markets, the accuracy and efficiency of job recommendation systems are crucial for Job seekers and employers. Traditional recommendation systems are based on keyword matching and then finding the similarity between job and CV vectors using some statistical method[1],[2]. The main problem with these methods is that they can easily capture and consider the quantifiable aspects of candidates and jobs during the matching. Still, they need help to correctly understand and match the contexts and semantics embedded within CVs and job descriptions[19],[20].

The advancements in natural language processing (NLP), especially the development of Large Language Models (LLMs) such as GPT-3 and GPT-4, come with a lot of opportunities for the improvement of these systems[9],[10] and [21]. LLMs can easily understand and generate human-like text, providing a more sophisticated analysis of CVs and job descriptions. These models leverage vast training data and advanced neural network architectures to capture text's embedded intricate patterns and relationships [4],[5].

High computational costs and the need for large datasets for fine-tuning are some challenges faced by LLMs. Furthermore, despite their capabilities, LLMs may still miss specific domain-specific nuances that traditional methods could capture [13],[14]. Therefore, a hybrid model leveraging the capabilities of both traditional as well as LLMs seems a potential option [15],[16].

In this paper, the authors propose a hybrid model that combines traditional methods with LLMs to enhance the accuracy and relevance of job recommendations.

The proposed methodology involves preprocessing CVs and job description databases and applying two distinct modules. The first module utilizes traditional techniques such as Term Frequency-Inverse Document Frequency (TF-IDF) and cosine similarity to compute similarity scores between CVs and job descriptions [11]. The second module uses the Large Language Model (GPT-4) to determine the similarity. This hybrid approach aims to address the limitations of existing job recommendation systems and provide a more comprehensive and effective solution for matching candidates with suitable job roles[12],[18].

II. LITERATURE REVIEW

Existing job recommendation systems are of five types: content-based recommendation systems, 2) knowledge-based systems, 3) collaborative-based systems, 4) Reciprocity-based systems, and 5) Hybrid-based systems, as shown below.

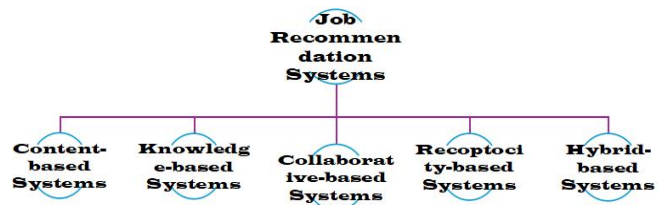


Fig. 1: Different types of Job Recommendation Systems

Content-based recommendation systems use the various attributes, such as skills, experience, and preferences, of jobs and job seekers to find the similarity between them [1],[4] and [11]. Knowledge-based systems use predefined rules and the explicit information provided by the user to recommend jobs, making them highly reliant on user input and domain knowledge [19]. Collaborative-based systems analyze user behavior and preferences and analyze patterns and trends among users to suggest jobs that other users with similar profiles have shown interest in [7]. Reciprocity-based systems are designed to account for mutual interest and fit, ensuring that recommendations benefit employers and job seekers. It considers factors like company culture, candidate aspirations, etc [2]. Lastly, hybrid-based systems combine elements from multiple recommendation approaches to improve accuracy and provide more comprehensive and personalized job suggestions [3],[8].

Year	Title	Type	Used features	Used Methodology
2007	Extending the applicability of recommender systems: A multilayer framework for matching human resources.[7]	Collaborative-based Recommendation	CV Data	• Multilayer framework for matching collaboration partners • Recommender-based approach to match human resources
2016	RésuméMatcher: A personalized résumé-job matching system. Expert Systems with Applications [6]	Content-based Recommendation	• Degree • Domain-specific skills • College • discipline	• Parsing and Tokenization • Semantic Labelling • Pattern matching • Measure skills similarity • Constructing an ontology of skills
2017	Text analysis for job matching quality improvement [1]	Content-based recommendation	• Job Features: workplace nearest station, start date, salary, position and tasks, requisite skills • Candidate Features: Personal information, skill and qualification, assessment results, work history, and previous working history with staffing company child care/elderly care priorities.	• text mining • Keywords are extracted by using KH Coder. • To identify meaningful keywords, experts were interviewed • Identified a number of important keywords indicating positive or negative influences on job matching • Frequency list of keywords • Hierarchical cluster analysis
2017	A hierarchical similarity-based job recommendation service framework for university students [2]	Reciprocity-based recommendation	• Time started hunting job • Time to get a job offer • Workplace nearest station • GPA • Student vote • Employer information	• Create students' profiles • Student similarity calculation • Job recommendation using weights
2017	Combining content-based and collaborative filtering for job recommendation system: A cost-sensitive Statistical Relational Learning approach. Knowledge-Based Systems [3]	Combining content-based and collaborative filtering	• Job Features: Job class, job skills, JobReqYear • Candidate Features: Class, city, skill, degree, most recent job title, last company, previous applied job	• Used Relational functional gradient boosting to learn features • Cost-sensitive learning with RFGB • Implemented using Regression Trees
2020	Job recommendation based on job profile clustering and job seeker behavior [8]	Hybrid-based Recommendation	• Job offers data • Job seekers interaction data • Ratings, likes, and reviews	• Data Collection • Textual processing of job offers • Job clustering • Matching and rating • Job recommendation
2020	Based on the application of AI technology in CV analysis and job recommendation [5]	Content-based Recommendation	• Education • Experience • Skill • Traits	• Competitive analysis • Personality trait analysis • Apply the DISC model • Term-frequency matrix • Similarity calculation • Job Recommendation
2022	AI-based suitability measurement and prediction between job description and job seeker profiles [4]	Content-based Recommendation	• Job Features: Job title, City, description, responsibilities, Education, Required skills • Candidate Features: Title, city, description, work experience, Education, Skills, certificates	• Linear Regression • Decision Tree • Adaboost • XGBoost
2023	JobRecoGPT- Explainable job recommendations using LLM [9]	LLM + Content-based recommendation	LLM extracts all features	Used LLM to extract features for job recommendation
2024	Enhancing job recommendation through llm-based generative adversarial networks.[10]	Content-based recommendation	Content in CV Content in interaction Relation history	LLM-Based Generative Adversarial Networks

Table 1: A study of existing Job Recommendation Systems

III. METHODOLOGY

The proposed job recommendation system employs a hybrid approach, integrating a Large Language Model (LLM) with the traditional model (based on TF-IDF and cosine similarity techniques) [11] to compute the similarity between CVs and job descriptions. This methodology ensures a robust and comprehensive job fit assessment, leveraging the strengths of both LLMs and Traditional models.

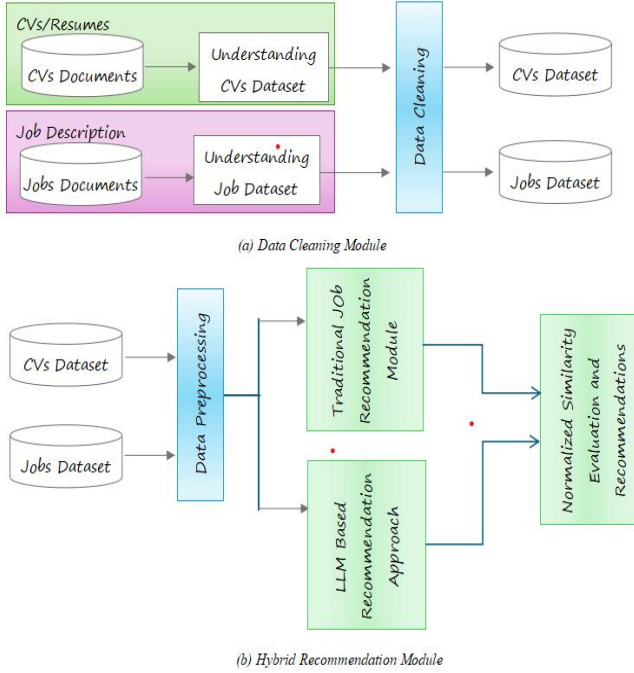


Fig. 2: A Hybrid Recommendation Model

A. Datasets: In this paper, the authors considered two datasets. Datasets regarding Jobs are taken from the Kaggle repository. In-house teams at PromptCloud and DataStock have generated the dataset. It contains 1 lakh unique records, each with ten features. The dataset features include Uniq Id, Crawl Timestamp, Job Title, Job Salary, Job Experience Required, Key Skills, Role Category, Location, Functional Area, and Industry. Another dataset is CVs, which consist of the CVs of many candidates with diverse skills and technologies. These CVs are in an unstructured format.

B. Methodology: First, the datasets with the CVs and the Job descriptions are cleaned and preprocessed to start recommendations. Here, the authors have primarily focused on the IT field, and the weightage process is used for the similarity. The recommendation between CVs and Job descriptions is based on the similarity score of various aspects like Programming languages, Frameworks, tools, certifications, experience, and soft skills[17]. The proposed weightage scheme is Programming Languages: 20%, Frameworks: 20%, Tools: 15%, Certifications: 10%, Experience: 25% and Soft Skills: 10%. Then, both different recommendation modules are applied to get their respective scores. Finally, the similarity scores are averaged to provide better Recommendations.

IV. EXPERIMENTS AND RESULTS

The section demonstrates the effectiveness of the hybrid approach. Here, the authors present the recommendations provided by this approach for one synthetic CV, i.e., CV1, as shown in Figure 3, against five synthetic job descriptions from JD1 to JD5, as shown in Figure 4. For demonstration purposes, all these CVs and Job Descriptions are LLM-generated.

```

Name: John Doe
Email: john.doe@example.com
Phone: 123-456-7890

Experience:
- Software Engineer at XYZ Corp (2018 - Present)
  - Developed and maintained web applications using JavaScript, React, and Node.js.
  - Led a team of 5 developers.
  - Implemented CI/CD pipelines using Jenkins.
  - Worked with Docker and Kubernetes for containerization and orchestration.
- Junior Developer at ABC Inc (2015 - 2018)
  - Assisted in developing e-commerce platforms using PHP and MySQL.
  - Participated in code reviews and pair programming.
  - Collaborated with designers to optimize user interfaces.

Education:
- B.S. in Computer Science, University of Somewhere (2011 - 2015)

Certifications:
- AWS Certified Solutions Architect
- Certified Scrum Master

Skills:
- Programming Languages: JavaScript, Python, Java, PHP
- Frameworks: React, Node.js, Express, Django
- Tools: Docker, Kubernetes, Jenkins, Git
- Soft Skills: Team Leadership, Communication, Problem-Solving

Projects:
- Real-time Chat Application using React and Node.js
- E-commerce Website with PHP and MySQL
  
```

Fig. 3. Unstructured Synthetic CV

```

JD1: Frontend Developer

We are looking for a Frontend Developer with experience in React and JavaScript. The candidate should be able to work with REST APIs and have a good understanding of version control systems like Git. Knowledge of HTML and CSS is a must.

JD2: Backend Developer

Seeking a Backend Developer skilled in Node.js and Express. The role involves working with databases like MongoDB and SQL. Experience with Docker and CI/CD tools is preferred.

JD3: DevOps Engineer

Hiring a DevOps Engineer with experience in Docker, Kubernetes, and Jenkins. The ideal candidate should have a strong background in CI/CD pipelines and cloud platforms such as AWS.

JD4: Full Stack Developer

Looking for a Full Stack Developer proficient in JavaScript, React, and Node.js. The role requires experience in both frontend and backend development, along with knowledge of databases and REST APIs.

JD5: Project Manager

We need a Project Manager with experience in agile methodologies and certifications like Scrum Master. The candidate should have excellent communication and leadership skills, and familiarity with software development processes.
  
```

Fig. 4. Unstructured Synthetic Job Descriptions (JDs)

As a next step, LLM converts the unstructured data to structured form. Figures 5 and 6 show the JSON format for the above CV and corresponding job descriptions.

```

{
  "Experience": [
    {
      "Role": "Software Engineer",
      "Company": "XYZ Corp",
      "Duration": "2018 - Present",
      "Responsibilities": [
        "Developed and maintained web applications using JavaScript, React, and Node.js",
        "Led a team of 5 developers",
        "Implemented CI/CD pipelines using Jenkins",
        "Worked with Docker and Kubernetes for containerization and orchestration"
      ]
    },
    {
      "Role": "Junior Developer",
      "Company": "ABC Inc",
      "Duration": "2015 - 2018",
      "Responsibilities": [
        "Assisted in developing e-commerce platforms using PHP and MySQL",
        "Participated in code reviews and pair programming",
        "Collaborated with designers to optimize user interfaces"
      ]
    }
  ],
  "Education": {
    "Degree": "B.S. in Computer Science",
    "Institution": "University of Somewhere",
    "Years": "2011 - 2015"
  },
  "Certifications": ["AWS Certified Solutions Architect", "Certified Scrum Master"],
  "Skills": {
    "Programming Languages": ["JavaScript", "Python", "Java", "PHP"],
    "Frameworks": ["React", "Node.js", "Express", "Django"],
    "Tools": ["Docker", "Kubernetes", "Jenkins", "Git"],
    "Soft Skills": ["Team Leadership", "Communication", "Problem-Solving"]
  },
  "Projects": [
    {
      "Name": "Real-time Chat Application",
      "Technologies": ["React", "Node.js"]
    },
    {
      "Name": "E-commerce Website",
      "Technologies": ["PHP", "MySQL"]
    }
  ]
}

```

Fig. 5. Structured Synthetic CV

```

Job 1: Frontend Developer:
{
  "Role": "Frontend Developer",
  "Responsibilities": ["Work with React and JavaScript", "Develop with REST APIs", "Use Vue.js"],
  "Required Skills": ["React", "JavaScript", "HTML", "CSS", "Git"],
  "Preferred Skills": ["Docker"],
  "Tools": []
}

Job 2: Backend Developer:
{
  "Role": "Backend Developer",
  "Responsibilities": ["Develop with Node.js and Express", "Work with database like MongoDB"],
  "Required Skills": ["Node.js", "Express", "MongoDB", "SQL"],
  "Preferred Skills": ["Docker"],
  "Tools": ["Docker", "CI/CD"]
}

Job 3: DevOps Engineer:
{
  "Role": "DevOps Engineer",
  "Responsibilities": ["Use Docker, Kubernetes, and Jenkins", "Develop CI/CD pipelines"],
  "Required Skills": ["Docker", "Kubernetes", "Jenkins"],
  "Preferred Skills": ["K8s"],
  "Tools": ["Docker", "Kubernetes", "Jenkins"]
}

Job 4: Full Stack Developer:
{
  "Role": "Full Stack Developer",
  "Responsibilities": ["Develop with JavaScript, React, and Node.js", "Work on frontend and backend"],
  "Required Skills": ["JavaScript", "React", "Node.js"],
  "Preferred Skills": ["Database", "REST APIs"],
  "Tools": []
}

Job 5: Project Manager:
{
  "Role": "Project Manager",
  "Responsibilities": ["Use agile methodologies", "Lead projects", "Work with software dev"],
  "Required Skills": ["Agile methodologies", "Leadership"],
  "Preferred Skills": ["Scrum"],
  "Tools": []
}

```

Fig. 6. Structured Synthetic JDs

LLM-based Recommendations: To leverage LLM's capability of comprehending natural language, while performing the experiment, no clear criteria were provided to recommend a job to a candidate or CV through an "unguided" approach. LLM was expected to find similarities

between the jobs and CVs through its own comprehension. Below, table 2 presents the similarity scores provided by LLM for CV1 against job descriptions from JD1 to JD5. The below table indicates that JD2 is the best match for CV1, followed by JD4, JD1, JD3, and JD5.

CVs	JDs	Similarity Score
CV1	JD2	.67
CV1	JD4	.65
CV1	JD1	.52
CV1	JD3	.50
CV1	JD5	.50

Table 2: LLM based Recommendations

Traditional Recommendations: The next section presents the recommendation results of the traditional model. It converts the CVs and job descriptions into TF-IDF vectors and finds their similarity scores.

Step 1: TF-IDF assigns weights to each term based on its frequency in the document and rarity across all documents. The following formulas are used for TF-IDF calculation [1],[11]:

Term Frequency (TF):

The term frequency of a term t in a document d is given by:

$$TF(t, d) = \frac{\text{Number of times term } t \text{ appears in document } d}{\text{Total number of terms in document } d}$$

Inverse Document Frequency (IDF):

The inverse document frequency of a term t across a set of documents D is given by:

$$IDF(t, D) = \log \left(\frac{\text{Total number of documents in the corpus } |D|}{\text{Number of documents containing term } t} \right)$$

TF-IDF:

The TF-IDF score of a term t in a document d is given by:

$$TF-IDF(t, d, D) = TF(t, d) \times IDF(t, D)$$

Step 2: Calculation for Cosine Similarity

This section calculates cosine similarity between the CV TF-IDF vector and each job description TF-IDF vector.

The cosine similarity between two vectors A and B (e.g., TF-IDF vectors of a CV and a job description) is calculated as follows:

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|} = \frac{\sum_{i=1}^n A_i B_i}{\sqrt{\sum_{i=1}^n A_i^2} \sqrt{\sum_{i=1}^n B_i^2}}$$

Here:

- $A \cdot B$ is the dot product of the vectors A and B .
- $\|A\|$ and $\|B\|$ are the magnitudes (Euclidean norms) of the vectors A and B .

CVs	JDs	Similarity Score
CV1	JD4	0.88
CV1	JD1	0.85
CV1	JD3	0.82
CV1	JD2	0.79
CV1	JD5	0.77

Table 3: Recommendations by traditional method

Final Similarity calculation:

Now, to find the final similarity scores, the following formula has been used:

$$\text{Similarity score (SS)} = (\text{SS1} + \text{SS2})/2$$

Where SS1: Similarity score from LLM-based model

SS2: Similarity Score from Traditional Recommendation System

CVs	JDs	Similarity Score
CV1	JD1	0.68
CV1	JD2	0.73
CV1	JD3	0.66
CV1	JD4	0.76
CV1	JD5	0.64

Table 4: Similarity Score from the hybrid approach

As per the above discussions, the job JD4, i.e., Full Stack Developer fits best to the given CV, followed by the JD2, i.e., Backend developer. The results demonstrate the higher accuracy of the hybrid model over the traditional and LLMs separately.

V. CONCLUSION AND FUTURE WORK

This research demonstrated the efficacy of a hybrid approach leveraging the traditional and Large Language Models. By combining, this model capitalized the quantitative rigor of traditional similarity metrics and the contextual understanding capabilities of LLMs by converting the unstructured CVs and Job descriptions into the structured format. The results demonstrated that the given CV was well fitted with the Full Stack Developer job, highlighting the importance of integrating multiple aspects such as programming languages, frameworks, tools, certifications, experience, and soft skills in the matching process. This hybrid methodology ensures a robust and comprehensive analysis and significantly improves accuracy.

As a future scope, dynamic weight adjustment algorithms, user feedback mechanisms, and real-time processing capabilities can be developed to create a highly personalized job recommendation system. This improved job recommendation system will undoubtedly help job seekers and employers.

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